CS & AI · EXPLAINED

How does a computer store a photo?

To a computer, your photo is just a grid of numbers.

THE HOOK

A photo looks smooth, real, full of colour.

But a computer doesn’t see a ”picture” at all — it sees a giant spreadsheet of numbers.

Let’s zoom in until you can see them. 🔍

TODAY'S QUESTION

Your selfie is colour and light. So how does a machine that only knows numbers hold it?

IDEA 1 / 4 — ZOOM IN FAR ENOUGH

A photo is a grid of tiny squares.

- Each little coloured square is a **pixel**.
- Zoom in far enough and you literally see the squares.
- A phone photo has **millions** of them.

pixel pixel pixel

Megapixels = millions of pixels. mega just means **million.**

IDEA 2 / 4 — WHAT ONE PIXEL IS

Every colour is just 3 numbers.

```
# one pixel = Red, Green, Blue (0–255)
( 255, 0, 0 )     → red
( 0, 0, 0 )       → black
( 255, 255, 255 ) → white
```

- Three lights: **Red**, **Green**, **Blue**.
- Each one a number from **0 to 255** — off to full brightness.
- Mix them and you get any colour you want.

IDEA 3 / 4 — WHY PHOTOS ARE "HEAVY"

So a photo = a huge list of numbers.

```
# one photo
millions of pixels  ×  3 numbers each  =  a giant list

stored as binary   # the same 0s and 1s, callback to EP05
```

- Every number is finally written as **0s and 1s**.
- More pixels → more numbers → a **bigger** file.
- That's why photos are measured in **MB**.

IDEA 4 / 4 — THE CLEVER SHRINK

Compression throws away what your eye won’t miss.

- Storing every single number is **wasteful**.
- JPEG drops tiny details you’d never notice…
- …and reuses repeated patterns instead of copying them.
- Result: far smaller than the raw image.

PNG VS JPEG

PNG keeps everything *(lossless)*.
JPEG drops detail *(lossy)*.

WHY IT MATTERS

This explains your whole gallery.

- **Megapixels** = how many pixels the camera captures.
- File **size** = how many numbers had to be stored.
- Zoom in too far or open an old, tiny photo → you see the squares: it looks **blocky / pixelated**.
- Sharing a logo? Pick **PNG**. A normal photo? **JPEG** saves space.

COMMON MYTH — "YEH GALAT SAMAJHTE HAIN"

✗ MYTH

"The computer stores the actual colours — the real picture is inside the file."

✓ REALITY

It stores **numbers** — the RGB values of each pixel. Your screen turns those numbers back into light. The "photo" only exists when it's **displayed**.

1. A photo = a grid of pixels, each just a few numbers (RGB).
2. Compression shrinks the file by cleverly dropping detail you won’t notice.

Your photo = a grid of numbers. 🖼️

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Next question: **What is "the cloud" actually?** ☁️